

Practical course

Compiler and Hardware Co-Design

(WiSe 2026/27)

Preliminary meeting

System Research Group

<https://dse.in.tum.de/>

Start at 15:00

Prerequisites

- Finish “Systems Programming” offered by our group beforehand
 - <https://github.com/ls1-sys-prog-course/docs>
- Please fill the form if you apply for this course
 - <https://collab.dvb.bayern/plugins/servlet/samlSso?redirectTo=%2Fpages%2Fviewpage.action%3FpageId%3D2752096784%26spaceKey%3DTUMcse%26title%3DPractical%252Bcourse%252BCompiler%252Band%252BHardware%252BCo-Design%252B--%252BWSS26%252B27>
- Please also apply to the matching system
 - <https://matching-in.cit.tum.de/>

About us

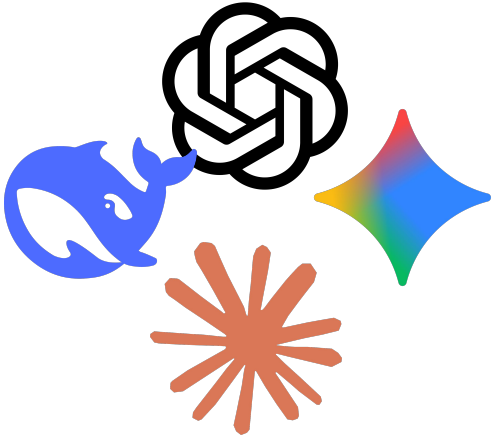
Systems Research Group

Prof. Pramod Bhatotia <https://dse.in.tum.de/>

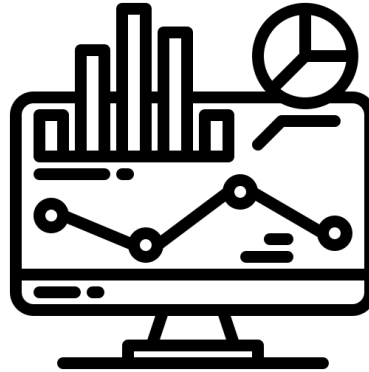
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Modern specialized workloads

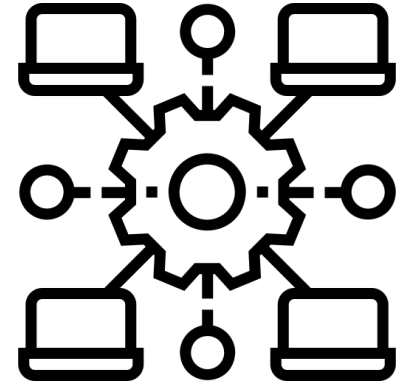
AI workloads



Data analytics

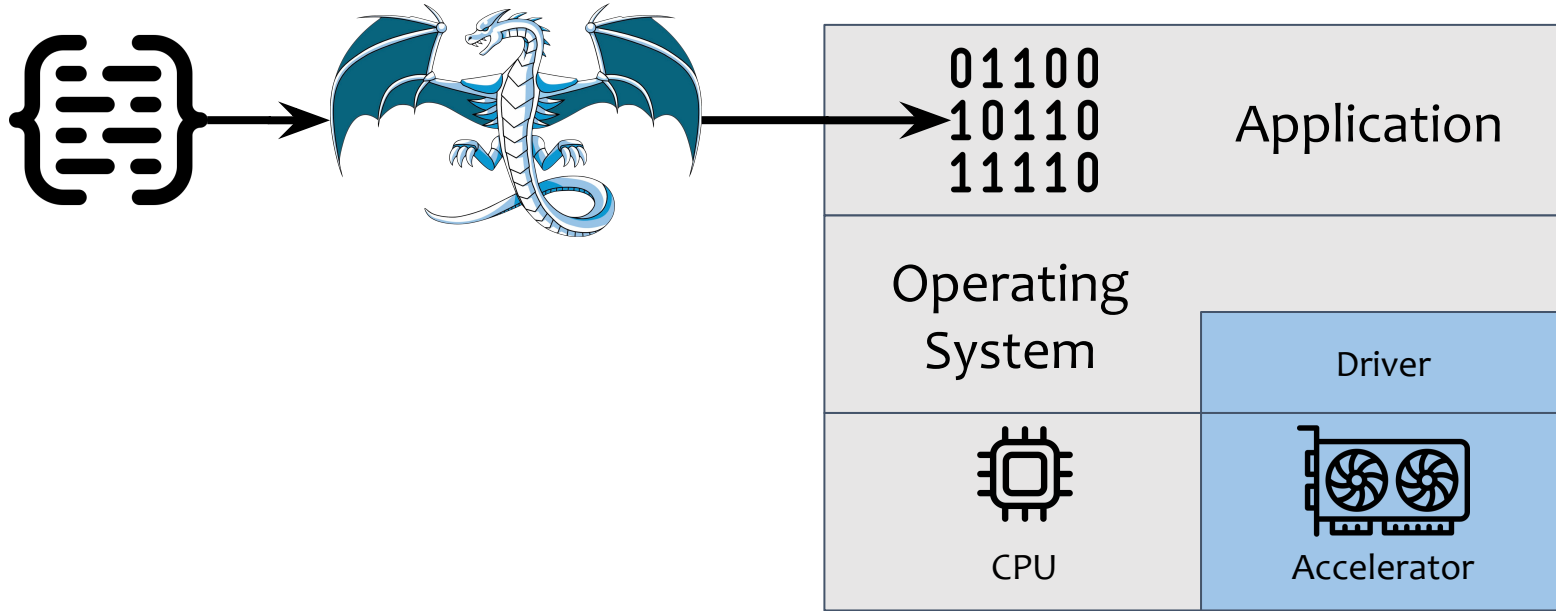


Networking, storage, and
real-time processing



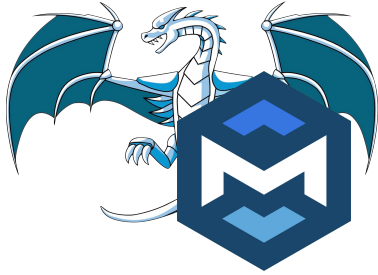
Modern applications run specialized workloads that need more performance and energy efficiency than CPUs alone can provide!

The modern accelerator stack



What you will build

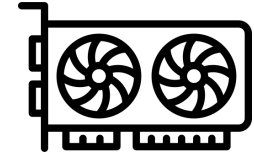
Compiler



Driver



Accelerator



A cross-layer solution to accelerate existing software

Prerequisites

- ***The students must successfully finish the following course beforehand.***
 - Practical Lab: Systems Programming (2023-)
 - Practical Lab: Advanced Systems Programming in C/Rust (offered in 2021-2023)
 - An equivalent course at another university (please check with the instructor!)

If the prerequisites are unclear, please check with the instructor!

Lab format

- Lab assignments
 - **3 practical programming exercises** with public & private unit tests
 - Deadline of 3-4 weeks depending on the difficulty/workload
 - Online submission
 - No further exam / quiz / projects
- Weekly meeting
 - Video with theoretical background coverage
 - Question and answer session to explain and discuss each assignment (in-person)
 - Slack channel for questions

Grading system

- Github classroom (<https://classroom.github.com/>) or similar
 - Template repository for each task with detailed instructions & test cases
- Automated tests
 - Points are distributed among the exercises based on the estimated workload (30-60 points)
 - Specially designed test cases with gradually increasing difficulty
 - Hidden tests to detect & prevent gaming the grading system

Languages / OS

- Languages
 - Depends on the task, C++/C/Verilog
 - Limited choice of allowed libraries (different per language)
 - Can be switched for each task
 - Some tasks require a specific language
- OS Environment information
 - All executables must run on **Linux, x86_64**
 - Use virtual machines if you run a different OS
 - You can also use the ITO VMs (lxhalle.in.tum.de)
 - You will run FPGA code in a simulator

Learning goals

- Learn how to build compiler optimizations using MLIR
- Understand how compiler passes, device drivers, and accelerator hardware fit together in one system
- Learn how to move computations from a general-purpose program to dedicated hardware.
- Gain hands-on experience with FPGA-based accelerator development and integration.
- Work with the full co-design flow from analysis and transformation to implementation and testing.

- Importantly, have fun!

Code of conduct

- University plagiarism policy
 - https://www.cit.tum.de/fileadmin/woobyx/cit/Studium/Im_Studium/Thesis/leitfaden-en_2016Jun22.pdf
- Decorum
 - Promote freedom of thoughts and open exchange of ideas
 - Cultivate dignity, understanding and mutual respect, and embrace diversity
 - Racism and bullying will not be tolerated
- **Please write your own code!**
 - The purpose is **to learn**, not to produce a software artifact (using AI)
 - You only cheat yourself

Interested?

 matching-in.cit.tum.de

 FAQs  Login 

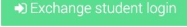
Matching platform

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Login for exchange students
(without TUM identifier)

 Exchange student login

Any questions? Visit the FAQs!

 FAQs

Please give feedback to matching@ds.in.tum.de · [Impressum](#) · [Datenschutz](#)

Fakultät für Informatik, Technische Universität München

- Martin Fink <mrtm.fink@tum.de>

We *strongly* prefer Zulip for all communications. Please join the Zulip channel.



Workspace: <https://zulip.in.tum.de>

Website: <https://github.com/compiler-hw-lab/docs>

Channel: [Compiler-HW-Co-Design](#) ¹⁴

Join us with TUM email address (@tum.de)